🔑 **Answer Key — Teacher Copy · Lesson 5-7**

MSTAR PRACTICE WORKSHEET

Lesson 5-7: Determine Surface Area of Prisms

Grade 6 Mathematics · Standard 6.GR.4

These are MSTAR-style questions written for this curriculum to rehearse the format. They are not official Maryland assessment items. MSTAR — Maryland's new state test, first given in spring 2027 — uses the same kinds of questions you see here: selected response, select-all, two-part evidence questions, and written responses.

Part A — Skills Check

One point each.

Question 1 SELECTED RESPONSE — CORRECT: C

What is the surface area of a rectangular prism with l = 5 cm, w = 4 cm, h = 3 cm?

A. 47 cm²

B. 60 cm²

C. 94 cm²

D. 94 cm³

SA = 2(5×4) + 2(5×3) + 2(4×3) = 40 + 30 + 24 = 94 cm².

Question 2 SELECTED RESPONSE — CORRECT: C

How many faces does a triangular prism have?

A. 3

B. 4

C. 5

D. 6

A triangular prism has 5 faces: 2 triangular bases and 3 rectangular lateral faces.

Question 3 SELECTED RESPONSE — CORRECT: C

A triangular prism has two triangular bases each with area 12 cm² and three rectangular faces with areas 30 cm², 40 cm², and 40 cm². What is the total surface area?

A) 110 cm²

B) 122 cm²

C) 134 cm²

D) 134 cm³

SA = 2(12) + 30 + 40 + 40 = 24 + 110 = 134 cm².

Question 4 SELECTED RESPONSE — CORRECT: C

A triangular prism has triangular bases with base 6 in and height 4 in. The prism is 10 in long and the two slant sides of the triangle are each 5 in. What is the surface area?

A) 124 in²

B) 160 in²

C) 184 in²

D) 184 in³

Triangular bases: 2 × (½ × 6 × 4) = 24 in². Three rectangular faces: 6(10) + 5(10) + 5(10) = 60 + 50 + 50 = 160 in². SA = 24 + 160 = 184 in².

Question 5 SELECTED RESPONSE — CORRECT: C

A rectangular time capsule box measures 3 in by 4 in by 5 in. How much wrapping paper covers its surface (no overlap)?

A) 47 in²

B) 60 in²

C) 94 in²

D) 94 in³

SA = 2(lw + lh + wh) = 2(3×4 + 3×5 + 4×5) = 2(12 + 15 + 20) = 2(47) = 94 in².

Question 6 SELECTED RESPONSE — CORRECT: C

A rectangular capsule crate is 6 cm long, 5 cm wide, and 4 cm tall. How much cardboard is needed to cover all its faces?

A. 74 cm²

B. 120 cm²

C. 148 cm²

D. 148 cm³

SA = 2(6×5) + 2(6×4) + 2(5×4) = 60 + 48 + 40 = 148 cm².

Part B — Test-Format Questions

Scoring notes and distractor rationales below each item.

Question 7 TWO-PART QUESTION

Part A — correct answer: D

A rectangular prism measures 6 cm long, 4 cm wide, and 2 cm tall. What is its surface area?

A. 24 square centimeters

B. 44 square centimeters

C. 48 square centimeters

D. 88 square centimeters

2(6×4) + 2(6×2) + 2(4×2) = 48 + 24 + 16 = 88 square centimeters.

- **A:** That is the VOLUME, 6 times 4 times 2, in cubic units. Surface area totals the six faces in square units.
- **B:** That counts one of each face instead of two. Every face has a matching opposite.
- **C:** That is the two largest faces only, 2 times 6 times 4. Four more faces remain.

Part B — correct answer: A

Why is surface area measured in square units while volume is measured in cubic units?

A. Surface area covers the outside faces, which are flat regions; volume fills the inside space.

B. Surface area is always smaller than volume.

C. Square units are used whenever a shape has straight sides.

D. Volume uses cubic units only for prisms.

Surface area is the total of flat face areas, so it uses square units. Volume measures how much space the solid fills, so it uses cubic units.

Question 8 WRITTEN RESPONSE · 2 POINTS

Type II Reasoning: Counting Only Three Faces

> Priya found the surface area of a box measuring 5 by 3 by 2 units. She wrote: '5×3 + 5×2 + 3×2 = 15 + 10 + 6 = 31 square units.'

Explain Priya's misconception and give the correct surface area.

> **Model answer:** Surface area = 2(15) + 2(10) + 2(6) = 62 square units. Priya found only three faces; every face has a matching opposite face, so each product is doubled.

2 points	Student identifies that a rectangular prism has 6 faces in 3 matching pairs, so each product must be doubled, and gives 62 square units.
1 point	Student gives 62 square units without explaining the matching pairs.
0 points	Student agrees with Priya or gives an unrelated response.

Part C — Show Your Thinking

Model answer and scoring guidance.

Question 9 WRITTEN RESPONSE · 2 POINTS

Compare two closed containers: a rectangular prism (8 × 6 × 4 in) and a triangular prism whose ends are right triangles with perpendicular legs 6 in and 8 in and hypotenuse 10 in. The triangular prism is 8 in long. Which uses less protective film to cover every face? Show your work.

> **Model answer:** The rectangular prism has surface area 2(8 × 6) + 2(8 × 4) + 2(6 × 4) = 208 in². The triangular prism has two ends totaling 2(½ × 6 × 8) = 48 in² and three rectangles totaling (6 + 8 + 10) × 8 = 192 in². Its total is 240 in². The rectangular prism uses 32 in² less film.

Award 2 points for a complete explanation with correct mathematics, 1 point for a correct answer with thin reasoning, 0 for an unrelated response.

Answer Key. Score written responses with the rubric and guidance above; award partial credit per the 1-point rows.